

# IoT based solar panel fault and maintenance detection using decision tree with light gradient boosting

P Sampurna Lakshmi<sup>a,\*</sup>, S Sivagamasundari<sup>a</sup>, Manjula Sri Rayudu<sup>b</sup>

<sup>a</sup> Department of Electronics and Instrumentation Engineering, Annamalai University, Chidambaram, Tamil Nadu, 608002, India

<sup>b</sup> Department of Electronics and Instrumentation Engineering, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad, Telangana, 500090, India

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## ABSTRACT

Solar PV (photovoltaic) technology has advanced greatly in recent years due to advantages such as renewability, environmental friendliness, simple maintenance, and dependability. Nevertheless, a number of PV faults may appear and result in degradation, a decrease in output power, or even a storm surge at different levels, depending on the outside working conditions and regular weather changes that might cause harm to the production, distribution, or setup, it is critical to monitor PVSs (PV systems) for their power generation efficiencies. IoT (Internet of Things) are evolving technologies that have been studied for enhanced fault detection and predictive analysis in the maintenance and environmental monitoring of solar power plants. This research work suggests a method based on MLTs (machine learning techniques) to analyze power data and predict faults for the maintenance of solar power plants. Input data from solar power plants consist of plant power generation and weather data which are first pre-processed and then trained using the suggested DT-LGB (Decision Trees with Light Gradient Boosting) algorithm to predict errors. The trained model was able to identify major/minor faults or anomalies present in input data. Conventionally these identifications require more effort in detection and maintenance. The results of this work showed that the suggested model obtained 8.74 MSEs ((Mean Square Errors), 2.96 RMSEs (Root Mean Square Errors), and  $R^2$  values of 0.9939 which is 12.8%, 6.8%, and 11.08% improved than the existing method respectively.

## 1. Introduction

Solar PV technology has evolved significantly in recent decades as an important source of renewable energy, mainly due to benefits like efficient energy generation, environment friendliness, ease of maintenance, and reliability. However, according to the outdoor working circumstances and periodic fluctuations in climatic conditions the possible damages associated with production, distribution, or setting up, numerous PV defects may emerge resulting in various levels of deterioration, reductions in output powers, or even storm surges. To overcome these issues, it is imperative to monitor the power generations of PVSs [1,2]. Most conventional methods incorporate manual examinations and remotely connected tracking and have several limitations including time consumption and complexity. IoTs have emerged as forefront technologies for examining the maintenance of PVSs and environmental monitoring with respect to demands in solar power plants for improved fault diagnostics and predictive analyses [3,4]. The IoT facilitates

communication and information sharing across a wide range of devices, systems, and services. Various studies have revealed that using IoT in the monitoring PVSs has several advantages, including better accuracy and efficiency, reduced human involvement, and hence lower costs. Furthermore, incorporating MLTs aids in large data points for electrical measurements, environmental data, or PV panel imaging [2,5].

IoT is playing a critical role in the day-to-day life of humans by empowering the connectivity of numerous physical devices done via the internet wherever the devices are perceptively accompanying together enabling novel kinds of communication among things and people and among things themselves to interchange the data for monitoring and controlling the devices from wherever about the globe using the internet linking [29]. Moreover, communication among machines or unlike devices is conceivable without human involvement using IoT applications [30]. The impression behind the plan IoT principle is to attach the sensors and devices of a special system on a common network through wired or wireless nodes. In general, IoT-based wireless systems are

\* Corresponding author.

E-mail addresses: [sampurnalakshmi\\_p@vnrvjiet.in](mailto:sampurnalakshmi_p@vnrvjiet.in) (S.L. P), [sivagamasundari67@gmail.com](mailto:sivagamasundari67@gmail.com) (S. S), [manjulasree\\_r@vnrvjiet.in](mailto:manjulasree_r@vnrvjiet.in) (M.S. Rayudu).

extensively selected in order to evade associated risks with wired systems. Whereas keeping in mind the desires of the near upcoming, where all devices will be smart, automated, and connected via the internet. For additional details, authors in [29] have examined some technical facts that debate to IoT-permitting technologies, protocols, and applications. They have enlightened the link among the IoT technology and the additional emerging technologies as well as cloud services, big data analytics and fog computing.

The persistence of PV monitoring systems is to suggest unceasingly a clear info about several parameters, namely the energy potential, extracted energy, fault detection, historical analysis of the plant, and associated energy loss. Besides, the monitored data can be used for defensive maintenance, primary detection of cautioning and evaluating the weather variations ... etc. [28,31]. Several classifications of PV monitoring systems created on the internet technology, data acquisition systems used and monitoring system methods have overviewed in detail in [28].

The availability of voluminous data and information in PVs, the use of data-driven technologies commonly known as AIs (Artificial Intelligences) for predicting performances of PVs are on the rise. Moreover, approaches based on MLTs have been presented. These models have been trained using existing information to anticipate and identify faults [6]. In recent years, models based on MLTs have emerged as popular choices in examining and identifying solar panel faults as they benefit from transfer learning feature that surpasses classic computer vision approaches. MLTs are powerful tools for not only classifying failures but also for understanding the process by which solar cell faults are recognised and identified. Such assessments enhance the dependability and endurance of solar panels, as well as assist in the management of their degradations and the enhancements of their performances. MLTs have the capability to analyze data without the need for pre-designed algorithms to extract features from input data [7]. The overall key contribution for the proposed method is to given as.

- Collect the solar dataset, which is initially pre-process to neglect the remove and avoid the unwanted information.
- Then pre-processed structure can give in to the proposed model to find the fault detection with the help of DT based Light Gradient Boosting.
- The expected outcomes like MSE, RMSE and  $R^2$  was calculated for performance evaluation.

The organization is as follows like the up coming section proceeds the existing methodologies with their problems then the succeeding portion describes about the recommended method. Afterwards the experimental results then over all conclusion for the article is mentioned.

## 2. Literature review

Solar Photovoltaic plants are being erected in large numbers across the globe at the moment, and these plants must be properly maintained and monitored on a continuous basis in order to remain safe and to sustain for longer periods. There are many different kinds of faults and failures that may occur in solar plants, and existing fault detection technologies are mostly utilized to protect and guard against certain problems like line-line, line-ground, arc and ground errors. Despite the existence of high universal standards (such as the IEC, NEC, and UL), undetected flaws endure to cause major difficulties in solar power plants [8]. There are several fault detection methods for the solar power plants accessible in the literature, each with a distinct level of accuracy, network provided, and algorithm intricacy. Estimations faults in PVs have been based on environment, climatic and satellite data. Moreover, few detection methods do not require any climatic data. An alternative strategy used is Electro Luminescence Images. Solar panels receive external excited currents through metal connections which act as light

emitting diodes. The photons emitted by this strategy which near wavelengths beyond 850 nm can be imaged using capable Si-CCDs cameras [9].

In recent times, smart systems combining AIs and the IOTs have been developed for monitoring, diagnostics and fault detections of PV solar power plants. The results significantly improved on deficiencies of present methodologies and marketed equipment's [10,11]. The solar PV panels are monitored and controlled using IoT nodes in smart monitoring systems. The earliest smart monitoring devices were created in Japan, and they included microprocessors, network radios, relays for connecting or obstructing panels, and sensors. These systems were simple and flexible for independent and grid-connected PV modules but the cost is also increased [12].

The advanced monitoring systems using IoT based technology, allowing the Solar PV plants to monitor its system performance and maintenance in real time, as well as to provide users with periodic updates on the health PVs. Furthermore, the usage of the IoT improves the comprehension of real-time operational characteristics. This facilitates access to the control of PVs located in distant places, as well as their effective and quick problem diagnosis and maintenance, as well as the copy of the power production and other performance data which helps in the investigation for further research [4]. An IoT-based STS solution that has been created and is being applied in solar panels in order to boost its efficiency using real-time wireless and internet communication characteristics [13]. An online remote monitoring system based on IoTs and fault diagnostics for PV arrays based on ELM (extreme learning machine) was proposed using data gateways and PVMC website [3,14]. The system gathered data successfully and sent them to data gateways, which records videos and obtains data locations which are uploaded to the PVMC website for monitoring. Studies have also proposed the use of sensor networks based on IoTs with the goal of creating advanced and trustworthy short-term forecasts of electricity generated by solar modules [4]. Usage of LSTM (Long Short-Term Memory) Networks based models assist in detecting deteriorations of PV module performances and designs based on Edge Computing for functioning and scalable operations have also been proposed but its complexity is the major issue [15].

Models of MLTs have emerged in recent years and have shown to be particularly successful in researching and recognizing faults in solar panels, since they benefit from a transfer learning characteristic that surpasses typical computer vision approaches. MLTs are useful not only for classifying failures but also for comprehending the process by which solar cell faults are identified. Such identification might increase the dependability and durability of solar panels, as well as aid regulate degradation and boost performance [7]. In a recent study the fuzzy logic-based diagnostic method that uses electrical parameters of crystalline silicon PV modules to get categorization values based on current-voltage measurements. This method helps in reducing the amount of money spent on PVs operational and as well as the amount of time spent on solar power systems [16]. Also, few fault detection algorithms make use of statistical analysis of various measurements of PVs, in addition system efficiency was reduced [17,18]. A suggested kernel-based ELM for defect detections in solar PV arrays showed good precision values with durability [19]. Power and voltage ratios were inputs for predicting errors in PVs where fuzzy theories and ANNs (artificial neural networks) were used resulting in detection accuracy of 92.1% [20]. Classifications based on Mamdani fuzzy logics on voltage and power ratio inputs with membership functions as singular inputs were examined where findings identified defects including partial shading and short-circuited PV modules in PVs [20,21]. In a recent study, the proposed eXplainable Fault Detection Systems model helped in understanding fault's causes and incipient fault detections with accuracy [22].

The major problems considered in the solar power plant fault detections are its high cost, unreliability, complexity of the circuit, high error value and less  $R^2$ . If any one of the factors is improved then the other factor performance will degrade, so overall performance was

affected. To overcome this problem, the recommended like DT-LGM was suggested. By the use of DT model will take the decision based to improve the quality of the system, that was based on the section LGM model.

### 3. Proposed model

This work's suggested model analyzes outputs of solar power plants and predict faults and maintenance requirements in these plants. The input power data was used to detect faults in panels and thereby train the model based on MLTs to predict future incident occurrences. Fig. 1 shows this work's proposed model. Inputs are first pre-processed and fed to the suggested DT-LGM model for learning which then predicts faults based on the training are illustrated in Fig. 2.

#### 3.1. Pre-processing of input data

The input data consists of two parts, the data generated form the plant and the weather data at the plant. From these two datasets, the input data columns namely

- Date time
- irradiation at the solar plant
- yield of the plant is extracted.

The following points are checked before sending the data for prediction.

1. Check for missing values
2. Merge the datasets by matching the input timeline

#### 3.2. DTs (Decision Trees) with light gradient boosting

The conventional DTs combined with the light gradient boosting algorithm to improve the overall prediction performance. This section presents proposed algorithm of DTs with light gradient boost.

##### 3.2.1. DTs

DTs almost mimic an individual's thinking for decisions, which make it easy to understand. The reasoning by DTs is straightforward since they have tree like structured. DTs, start with the root nodes and move towards forecasts of dataset categories. By comparing values of records (actual dataset) attributes with root attributes, DTs follow branches and move to next nodes when algorithm double-checks attribute values with

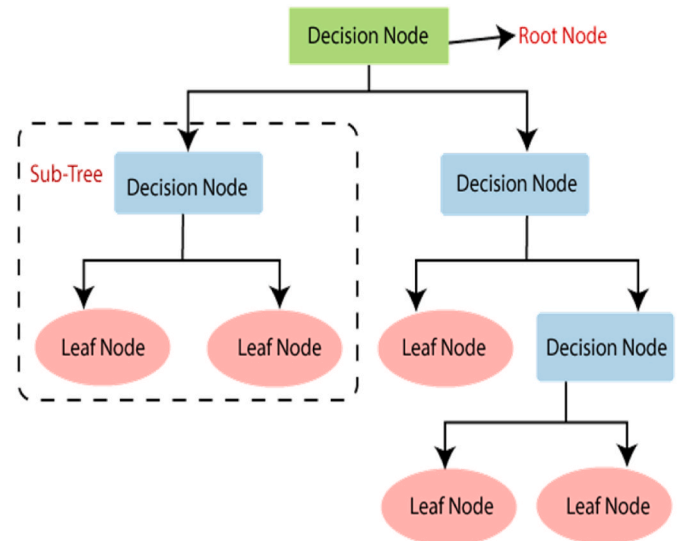


Fig. 2. DT structure.

other sub-nodes. The processes are iterated until leaf nodes of the trees are reached.

- Root node: The starting node of the DTs.
- Leaf node: The final node without any branch.
- Decision node: Intermediate nodes between root node and leaf node.
- Splitting: The phenomena that occurs after the decision node.
- Sub-Tree: The branch network of the DTs.

These terms are illustrated with the help of Fig. 3. The algorithmic DTs is listed below:

#### Algorithm 1. Decision Trees

- 
- Step 1:** Read the input data features at the root node.  
**Step 2:** Convert the root node into a decision node and add a condition to the decision node which results in two leaf nodes.  
**Step 3:** Select the left leaf node and transform it into a decision node by adding a condition. Repeat the same thing with the right node.  
**Step 4:** Perform step 3 repeatedly by updating the conditions in the decision nodes until the leaf nodes classify the input data.  
**Step 5:** Save the network structure with the conditions in the decision nodes.
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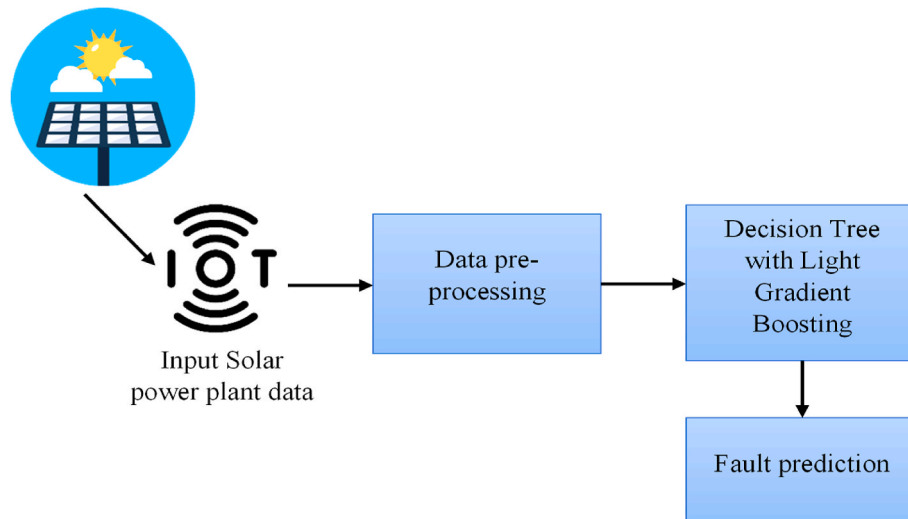


Fig. 1. Proposed Fault detection model.

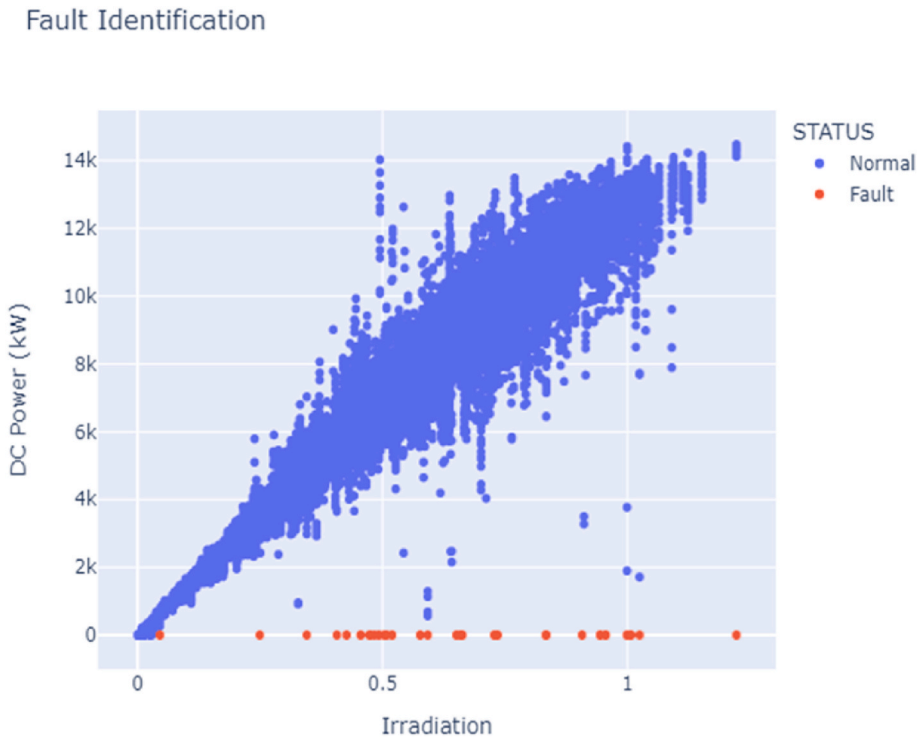


Fig. 3. Fault identification.

### 3.2.2. LGB (light gradient boosts)

LGB frameworks use tree-based learning methods are very potent algorithm in computations. They are regarded as quick-processing algorithms. Light GBM (Light Gradient Boosting Machine) methods grow vertically which implies they develop as leaves as opposed to other algorithmic trees which grow horizontally. Light GBM selects leaves with significant losses for growth. While expanding leaves, they can minimize losses more than level-based strategies. In contrast to prior boosting algorithms that develop trees at level-by-level basis, Light GBM divides trees into leaves. They select leaves with greatest delta losses for growth which results in reduced losses when compared to level-based techniques since leaves are fixed. Leaf-wise tree growths might increase complexities of models, leading to over fits of data while using small datasets.

### 3.2.3. Proposed DTs with light gradient boosting

The proposed DT-LGB is designed to have advantages over conventional methods like:

- Reduced training time
- Reduced memory usage
- Higher accuracy
- Supports large databases

The detailed algorithm is described here.

#### Step 1: Select a value from the input distribution.

This step is used to provide an initial point for the algorithm to start the training process. Most frequently, the average value of the distribution is calculated and the average value is used to begin the algorithm.

#### Step 2: Calculate the Residue for each point in the distribution.

In this step, the difference between the actual value and predicted value is calculated. This difference is termed as residue. At this stage, the

predicted value becomes the average value calculated in step 1.

#### Step 3: Construct the DTs

In this step, the DTs are constructed with root nodes, decision nodes and leaf node.

#### Step 4: Predicting targetted labels using trees

Each sample is processed until it reaches a particular lead, at which time it travels through the decision nodes of the freshly built tree. The value may be predicted based on the residual in the aforementioned leaf.

#### Step 5: Calculate the new residue values

The predicted values are subtracted from the actual values to get the updated residues.

#### Step 6: Boost the nodes with higher residues

Boosting is an ensemble learning technique where occurrences that are challenging to categorize are given larger weights, suggesting that future learners can concentrate more on studying incorrectly categorized data instances. The final model would be a weighted average of  $n$  different instances which have issues.

#### Step7: Repeat Steps 3, 4, 5, and 6 till the residue values are minimum and the desired prediction accuracy is achieved.

Gradient Based One Side Sampling is an innovative method employed in the suggested model that reduces the samples on the basis of gradients. Problems with modest gradients have received adequate training (a relatively small amount of training error), whereas instances with big gradients have received inadequate training. It is possible to down sample in a naïve manner by ignoring cases with small gradients and concentrating entirely on examples with big gradients; however,

doing so would result in an alteration of the data distribution. In a nutshell, GOSS chooses to keep instances that have big gradients while randomly picking examples that have modest gradients.

It is possible to accelerate the process of feature tree learning if one can down sample the feature. DT with Light GBM is able to do this by combining many functionalities into a single package. In most cases, we work with data that has a large dimensionality. These data contain numerous characteristics that are incompatible with one another, meaning that they can never both take the value zero at the same time. LightGBM is able to recognize these features without risk and combine them into a single feature in order to minimize the complexity.

### 3.3. Fault prediction

In today's world, one of the most important tools for keeping solar photovoltaic plants operating at a high level of technical and economic performance over the long term is the execution of correct and methodical preventive maintenance methods. Analytical monitoring systems have been established all over the world in order to detect potential problems in a timely manner through the evaluation of the performance of PVs. Statistical methods based on data mining and MLTs have recently come to light as promising methods for both fault predictions and early detections. It is challenging because of voluminous pertinent data generated in PV systems and modelling numerous intricate PV plant components. Specifically, recent monitoring operations of PVs have recently used important enabling technologies and paradigms, most notably environments for IoTs and MLTs to handle huge amounts of data.

The following is a list of the most typical issues associated with panels:

- **The panels have certain trouble spots:** The areas of the panels that are overloaded and, as a result, get warm are referred to as hot spots. On solar panels, hotspots are almost often the consequence of poorly soldered associates or a flaw in the physical composition of the solar cells themselves. Inadequately soldered connections may be roots of low resistances of panels that receive power from cells resulting in voltage may increases and thus forming hotspots at soldered spots or in cells. This condition lowers performances of PV panels and shortens their lifespan. They may eventually lead to short circuits.
- **Micro-cracks:** It is a phenomenon that occurs frequently in crystalline photovoltaic panels, and it is one that we have seen. These miniscule rips in the solar cells are so small that they are almost impossible to see. It is possible for PV modules to develop micro-cracks not only during the manufacturing process, but also during shipment or as a result of improper handling practices during installation. In a similar vein, microcracks may not necessarily result in a loss of output right away but may instead get larger completed times, for example as a result of thermal strain or as a result of the influence of climatical and seasonal factors. If the microcracks are significant enough, the solar cells will be harmed, which will reduce output. The quantity of energy that is produced by the solar cell will be significantly impacted by damage incurred at the contact sites.
- **Snail trail contamination:** This is a discoloration issue with the panel that normally does not show up until a couple of years after production started. When making solar cells, faulty front metallization silver paste is occasionally used which might be one of the factors contributing to the formation of snail trails. However, this is not the only one. A faulty silver paste can cause the panel to become damp, and when this happens, an oxidation reaction can take place between the silver paste and an capturing substance known as EVA (ethylene vinyl acetate).
- **PID Effect:** "Potential Induced Degradation" is the abbreviation for this phenomenon. If there is a potential difference among the panel and the earthing, this problematic can develop obvious. The solar panel is earthed for protection reasons, nevertheless doing so may

cause a possibly deadly potential difference among the earthing and the voltage the panel produces. In certain cases, this generates a voltage that is somewhat discharged over the primary power circuit. As a straight result of this effect, the PV panel's accomplishment will remain to decline with time, and its age will increase more rapidly. PID has been seen to have a negative impact on solar plants, resulting in a performance loss of up to 10%. At this time, we are looking into a number of different methods that may potentially allow us to reverse the PID effect.

- **Internal corrosion and delamination:** Internal corrosion, sometimes referred to as rusting, takes place when moisture enters the panel and makes it wet. The panels must also be waterproof and airtight. To organize this, the panel's constituent parts—the glass layer, the solar cells, and the back sheet—are coated while operating in a vacuum. However, if the Coating process was not done appropriately or was shortened, this might consequence in delamination throughout the process. The separation of the components that were covered is known as delamination. Bubbles or moisture seepage can be caused by delamination, but it's also possible for the module trim to have been installed improperly, for example. Corrosion, which may be identified on the panel as darker areas due to the presence of moisture, is caused by dampness.

Irrespective of the type, when faults occur in panels, output powers drop. The logic used to predict the faults in the panels is based on the predicted power value and the obtained power value. After the model is trained with the input data, new upcoming power values are predicted. Whenever the obtained power values are less than the predicted power value under normal radiation circumstances, a fault is said to have occurred. This way, the faults are predicted beforehand making it easy for the maintenance team to fix the issues.

## 4. Experimental results

The data was extracted over a 34-day period and consists of two sets of data regarding two solar power facilities in India and for training as well as testing was splitted in the ratio of 80:20, the details of data set was as:

- information about electricity generation (collected from an inverter connected to an array of solar panels)
- weather data (collected from the sensors array placed on the solar panels)

The input data consists of information like date times, DC/AC powers, total yields, ambient/module temperatures, irradiances etc.

Fig. 3 shows the fault identification plot in the solar power plant. The implementation was evaluated by the use of JAVA script. The X-axis represents the radiation on the solar panel. The Y-axis represents the DC power output. The Plot contains blue dots representing normal operation and red dots indicate the occurred faults.

The power forecast made by the suggested model is shown in Fig. 4. The output DC power is shown by the Y-axis, whereas the time is characterized by the X-axis. The blue line on the graph represents the actual measured DC power at the power plant. The projected DC power value is represented by the red line. The difference among the real DC power output and the predictable DC power output is revealed by the green line. The DC power increased after 6 a.m. and reached the peak value at around 10 a.m. The power output was considerably high till 3 p.m. The obtained maximum power output is around 13,500 kW. The power slowly reduced after 3 p.m. till 7 p.m. This figure denotes a normal operation where no faults occurred during the day. The proposed method predicted the output DC power value with high accuracy.

Fig. 5 shows the scenario where faults have occurred. Similar to Fig. 4, the X-axis signifies the time and the Y-axis characterizes the output DC power. The blue line indicates the actual measured DC power

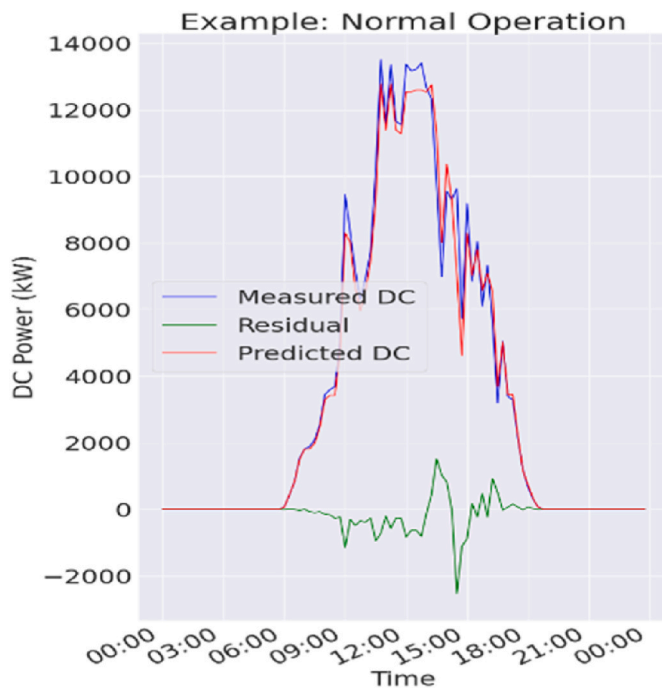


Fig. 4. Power prediction by the proposed model.

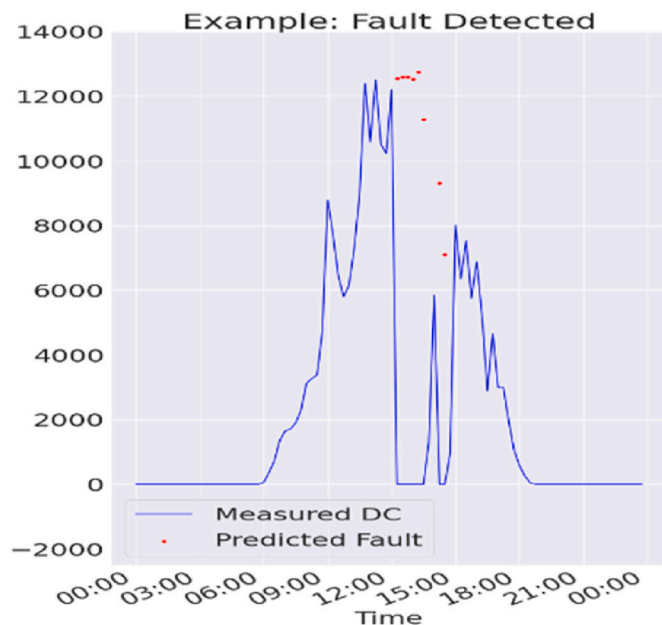


Fig. 5. Fault prediction.

obtained. The Red dots in the plot show the predicted faults occurred. The faults have occurred in the plant between the duration of 12 p.m. to 3 p.m. As occurrence of faults reduces the power output, thus the blue line in the graph dropped to zero. The suggested model predicted faults accurately which were indicated in the red dots.

Table 1 compares the recommended model with the practises currently in use. The following were variables employed in the comparison analysis were MSEs, RMSEs and  $R^2$ . Linear Regression obtained an MSEs of 16.47, RMSEs of 4.057900 and  $R^2$  of 0.988566365029754. Non-Linear Model obtained an MSEs of 14.36, RMSEs of 3.79 and  $R^2$  of 0.9900284653252387. Polynomial Regression produced an MSEs of 14.21, RMSE of 3.7 and  $R^2$  of 0.9901326856654404. Support Vector

Table 1  
Comparative analysis.

|                                  | MSEs        | RMSEs       | $R^2$                     |
|----------------------------------|-------------|-------------|---------------------------|
| Linear Regression [23]           | 16.47       | 4.057900    | 0.988566365029754         |
| Non-Linear Model [24]            | 14.36       | 3.79        | 0.9900284653252387        |
| Polynomial Regression [25]       | 14.21       | 3.77        | 0.9901326856654404        |
| Support Vector Machine [26]      | 390.86      | 19.77       | 0.72860447659764          |
| Gradient Boosting Regressor [27] | 10.21       | 3.20        | 0.9927148347330779        |
| <b>Proposed model</b>            | <b>8.90</b> | <b>2.98</b> | <b>0.9938168032244021</b> |

Machine produced an MSEs of 390.86, RMSEs of 19.77 and  $R^2$  of 0.72860447659764. The Gradient Boosting Regressor algorithm obtained an MSEs of 10.21, RMSEs of 3.20 and  $R^2$  of 0.9927148347330779. The suggested model obtained an MSEs of 8.9, RMSEs of 2.98 and  $R^2$  of 0.9938168032244021 which are the best compared to the existing models.

## 5. Conclusion

The use of solar cell panels as an effective power source for the creation of energy has been explored for a very long time. Any kind of damage to the surface of the solar panel will result in a loss of a generation of power and a lower yield. Defects are created by mechanical and chemical environmental forces that stress the panel when it is functioning in the field. These natural causes include snow, sun, wind, and extreme cold. This research proposes strategy based on MLTs to analyze power data and forecast defects and maintenance needs in solar power facilities. The input data from the solar power plant, collected with the help of IoT, is first pre-processed, and then it is trained using the suggested DT-LGB method. This is done in order to forecast the defects that are present in the input data. The output power will decrease if there is an issue with the panel. The reasoning that is utilized to forecast defects in the panels is based on the power value that was predicted and the power value that was actually attained. A fault is considered to have occurred when the measured power values that were obtained are lower than the expected power value that would be achieved under normal radiation conditions. In this manner, the failures may be foreseen in advance, which makes it much simpler for the maintenance crew to resolve the difficulties. The trained model is able to detect faults that are present in the panel. This is a significant improvement over the traditional method, which requires significantly more work to detect faults and provide maintenance. The findings indicate that the model under consideration obtained an MSEs value of 8.74, RSMs value of 2.96, and  $R^2$  value of 0.9939 which is 12.8%, 6.8% and 11.08% improved than the existing method respectively. In future the size of the data set can be used for real-time applications.

## CRedit authorship contribution statement

**P Sampurna lakshmi:** Conception and design of study, Revising the manuscript critically for important intellectual content, Approval of the version of the manuscript to be published, Conceptualization, Writing – review & editing. **S Sivagama Sundari:** Acquisition of data, Drafting the manuscript, Approval of the version of the manuscript to be published, Funding acquisition, Writing – original draft, Data curation. **Manjula sri R:** Acquisition of data, Revising the manuscript critically for important intellectual content, Analysis and/or interpretation of data, Approval of the version of the manuscript to be published, Funding acquisition, Writing – review & editing, Formal analysis, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

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